# RTL8C (A6ZKI3): A Domesticated Capsid-Derived Proteostasis Adaptor Coupling PEG10 and UBQLN2

**Gene:** RTL8C (synonyms: CXX1, FAM127A, MAR8, MAR8C, MART8C) **UniProt:** A6ZKI3 · **HGNC:** HGNC:2569 · **Organism:** *Homo sapiens* (human) **Protein family:** FAM127 family · **Domain:** RTL1/1-8/LDOC capsid-like (InterPro IPR032549; Pfam PF16297)

## Summary

**RTL8C** (UniProt **A6ZKI3**; gene synonyms **CXX1**, **FAM127A**, **MAR8**, **MAR8C**, **MART8C**) is a small (~13 kDa, 113-amino-acid) human protein that is **not an enzyme, transporter, or classical signaling molecule**, but rather a **protein–protein interaction adaptor/regulator** derived from an ancient domesticated retroelement. It belongs to the FAM127 family and carries a single retroviral **Gag/capsid-like (CA) domain** (InterPro IPR032549; Pfam PF16297, "RTL1-8_LDOC"). In essence, RTL8C is a truncated structural homolog of the N-terminal capsid domain of PEG10, another domesticated Ty3/Gypsy LTR-retrotransposon-derived gene. RTL8C is one of three nearly identical human paralogs — **RTL8A/B/C** (also called CXX1a/b/c or Mart8a/b/c) — encoded on the X chromosome, and functional findings for "RTL8" apply across this cluster given >92% sequence identity.

The primary molecular function of RTL8C is to act as a **capsid-domain adaptor at the interface of endogenous-retroelement particle biology and cellular proteostasis**. It performs two linked, experimentally supported roles. First, it binds the **N-terminal capsid domain of PEG10** and **antagonizes PEG10 virus-like particle (VLP) assembly/release** — RTL8 cannot form particles on its own but is packaged into PEG10 particles and, when co-expressed, suppresses their formation while raising intracellular PEG10. Second, it is an **obligate bridge for the proteasomal shuttle factor UBQLN2**: RTL8 localizes to **nucleoli**, drives the **nuclear translocation of UBQLN2** into ubiquitin-enriched subnuclear protein-quality-control (PQC) compartments, and — critically — is **required for UBQLN2 to complex with PEG10 and be recruited to cytoplasmic stress granules** under stress.

Thus RTL8C functions in **two subcellular locations** (the nucleolus/subnuclear PQC bodies and the cytoplasmic stress granule) and operates within the **ubiquitin–proteasome system and stress-granule proteostasis pathways**, with direct relevance to UBQLN2/PEG10-linked neurodegeneration (ALS/FTD). The UBQLN2–RTL8 relationship is reciprocal: UBQLN2, which normally directs substrates to degradation, instead **stabilizes** RTL8/CXX1 protein levels. This report synthesizes six confirmed findings drawn from primary experimental, proteomic, and structural/evolutionary evidence into a coherent mechanistic model.

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## Key Findings

### Finding 1 — RTL8C is a domesticated retroelement-derived gene encoding a small Gag/capsid-like protein

RTL8C is a product of **retroelement domestication**. The gene belongs to the **SIRH/Mart family** — "Mammalian retrotransposon transcripts (Mart1–9), also called sushi-ichi-related retrotransposon-homolog (SIRH) genes," which are "a class of Ty3/gypsy LTR-retroelements showing moderate homology to the sushi-ichi LTR-retrotransposon in pufferfish" (PMID: 25888968). RTL8C corresponds to **Mart8c/CXX1c** within this family. These genes arose when LTR-retrotransposon coding sequences were captured and repurposed ("domesticated") by the mammalian genome during evolution.

Structurally, RTL8 is a **minimal capsid module**. It "resembles a truncated form of PEG10 that shares homology with the N-terminal gag-like capsid domain" (PMID: 39775359). Where PEG10 (Mart2) retains both the Gag-like capsid (CA) domain and a Pol-like region, RTL8 has been reduced to essentially just the **N-terminal CA domain** — a ~13 kDa protein. This architecture is the molecular basis for its function: it provides a capsid-interaction surface without the machinery to independently build a particle.

RTL8C is one of **three nearly identical paralogs** (RTL8A/B/C = CXX1a/b/c = Mart8a/b/c) clustered on the X chromosome. Because the human reference annotations and the experimental literature frequently refer collectively to "RTL8," the functional characterizations below are properly understood as applying to the RTL8 cluster, of which RTL8C is a member.

## Finding 2 — RTL8 binds the PEG10 capsid and antagonizes PEG10 virus-like particle formation

The most direct evidence for RTL8's primary molecular function comes from Campodonico et al. 2024 (PMID: 39775359), which defined RTL8 as a **negative regulator of PEG10 particle assembly**. The key experimental result: "Alone, RTL8 is unable to form VLPs, but was incorporated into PEG10-derived particles. RTL8 co-expression decreased the abundance of PEG10 VLPs and increased intracellular levels of PEG10, suggesting a model where RTL8 inhibits PEG10 VLP formation or release."

The mechanism is a **capsid–capsid interaction**: "RTL8 bound to the N-terminal domain of PEG10 capsid, and modulation of RTL8 influenced PEG10-derived VLP abundance in naturally producing cells" (PMID: 39775359). Because RTL8 is itself a truncated capsid domain, it can engage PEG10's capsid-assembly interface and act as a **dominant-negative / poison-subunit-like antagonist**: it gets incorporated into nascent particles but, lacking the full capsid-plus-Pol architecture, it disrupts productive assembly or release. The net effect is fewer secreted VLPs and more PEG10 retained inside the cell.

This interaction is physiologically plausible because RTL8 is "broadly expressed in many of the same tissues as PEG10, including human brain," placing the antagonist and its target in the same cells. This finding establishes RTL8's role as a **stoichiometric regulator of a domesticated retroviral capsid**, and is central to the proteostasis functions described below, since PEG10 particle behavior is tied to stress-granule biology.

## Finding 3 — RTL8 localizes to nucleoli and drives UBQLN2 into subnuclear protein-quality-control compartments

RTL8's second major role is as a **spatial regulator of UBQLN2**, the proteasomal shuttle factor mutated in familial ALS/FTD. Mohan et al. 2022 (PMID: 35247097) identified RTL8 in a mass-spectrometry screen for UBQLN2 interactors and confirmed the **UBQLN2–RTL8 interaction both in vitro** (with recombinant proteins) **and in vivo** (in mouse brain), establishing it as a direct, physiologically relevant interaction rather than a high-throughput artifact.

The localization data pinpoint where RTL8 acts: "Under endogenous and overexpressed conditions, RTL8 localizes to nucleoli. When co-expressed with UBQLN2, RTL8 promotes nuclear translocation of UBQLN2" (PMID: 35247097). RTL8 also facilitated UBQLN2 nuclear entry **during heat shock**, indicating the function is engaged under proteotoxic stress. Once in

the nucleus, "UBQLN2 and RTL8 colocalize within ubiquitin-enriched subnuclear structures containing PQC components" (PMID: 35247097), directly tying RTL8 to **nuclear protein quality control**.

Two specificity details reinforce the mechanistic precision of this finding. The effect was **specific to UBQLN2** and did not extend to the related ubiquilins UBQLN1 or UBQLN4, arguing that RTL8 recognizes a particular feature of UBQLN2. And the relationship is **reciprocal**: UBQLN2 stabilizes RTL8 protein levels (expanded in Finding 5). Together these data cast RTL8 as a **targeting/chaperone-like adaptor** that controls the subcellular distribution of a proteostasis machine.

## Finding 4 — RTL8 is the obligate bridge coupling UBQLN2 to PEG10 and to stress granules

The two arms of RTL8 biology — PEG10 capsid binding and UBQLN2 regulation — converge in the cytoplasmic stress-granule compartment. Mohan et al. 2025 (PMID: 40680123; see also the companion report PMID: 39484508) showed that "UBQLN2 is regulated by the linked actions of two retrovirus-like proteins, RTL8 and PEG10." Crucially, "**RTL8 confers on UBQLN2 the ability to complex with and regulate PEG10. PEG10, a core component of stress granules, drives the recruitment of UBQLN2 to stress granules under various stress conditions but can only do so when RTL8 is present**" (PMID: 40680123).

This establishes RTL8 as a **non-redundant, obligate molecular bridge**: without RTL8, UBQLN2 cannot engage PEG10, and PEG10 cannot pull UBQLN2 into stress granules. RTL8 therefore sits at the hub of a three-protein module (RTL8–UBQLN2–PEG10) that governs how a proteasomal shuttle factor is distributed into stress-induced biomolecular condensates.

Functionally, perturbing the module has downstream consequences for condensate dynamics: changes in UBQLN2, RTL8, or PEG10 levels **remodel stress-granule disassembly kinetics and translation recovery**, and within stress granules PEG10 forms virus-like particles — "underscoring the structural heterogeneity of this class of biomolecular condensates" (PMID: 40680123). This connects RTL8's capsid-binding activity (Finding 2) to the physiology of stress granules, and links the domesticated-retroelement system to the broader proteostasis stress response.

## Finding 5 — UBQLN2 reciprocally stabilizes RTL8/CXX1, protecting the Gag-like protein from degradation

An independent, unbiased line of evidence corroborates the tight coupling between UBQLN2 and the RTL8 cluster. Whiteley et al. 2021 (PMID: 33277362) performed **global proteomics on Ubqln2-based murine ALS models and Ubqln2-null mice** and found that "among the specific proteins whose abundance is linked to UBQLN2 function, the strongest hits were the ubiquitin ligase TRIM32 and two retroelement-derived proteins, PEG10 and CXX1B" (CXX1B being an RTL8 paralog).

The direction of the effect is notable. UBQLN2 is a shuttle factor that typically **directs substrates to the proteasome for degradation**, yet in this proteomic analysis it instead **conferred strong protection from degradation** on the Gag-like RTL8/CXX1 protein, stabilizing rather than destroying it. This independently reproduces the reciprocal stabilization reported by Mohan et al. 2022 (Finding 3) and demonstrates that the RTL8–UBQLN2 relationship is a genuine, mutually stabilizing partnership rather than a simple substrate–enzyme interaction. The co-identification of **both PEG10 and an RTL8 paralog** as top UBQLN2-linked proteins in a hypothesis-free screen is strong orthogonal support for the integrated RTL8–UBQLN2–PEG10 module.

## Finding 6 — RTL8C folds as a compact capsid-like domain and is ~93–96% identical to its paralogs

Structural and evolutionary bioinformatics confirm that RTL8C is a **single, well-folded capsid-like domain** and that the three paralogs are functionally interchangeable. The AlphaFold model for A6ZKI3 is predicted with **high confidence** (mean pLDDT 89.4; 87% of residues ≥70, 72% ≥90, only 2% <50). The central region (residues 21–95) forms a **well-ordered globular core** (mean pLDDT 95.4, "very high"), flanked by modestly less ordered N- (residues 1–20, pLDDT 73.2) and C-termini (residues 96–113, pLDDT 82.5). This profile is consistent with a **single folded Gag/capsid-like (CA) domain** — not an intrinsically disordered protein and not a multidomain protein — supporting the interpretation that RTL8C's function is mediated by a compact structured capsid surface.

Pairwise comparison of the three human paralogs (RTL8C/A6ZKI3, RTL8A/Q9BWD3, RTL8B/Q17RB0; all 113 aa) shows they are nearly identical:

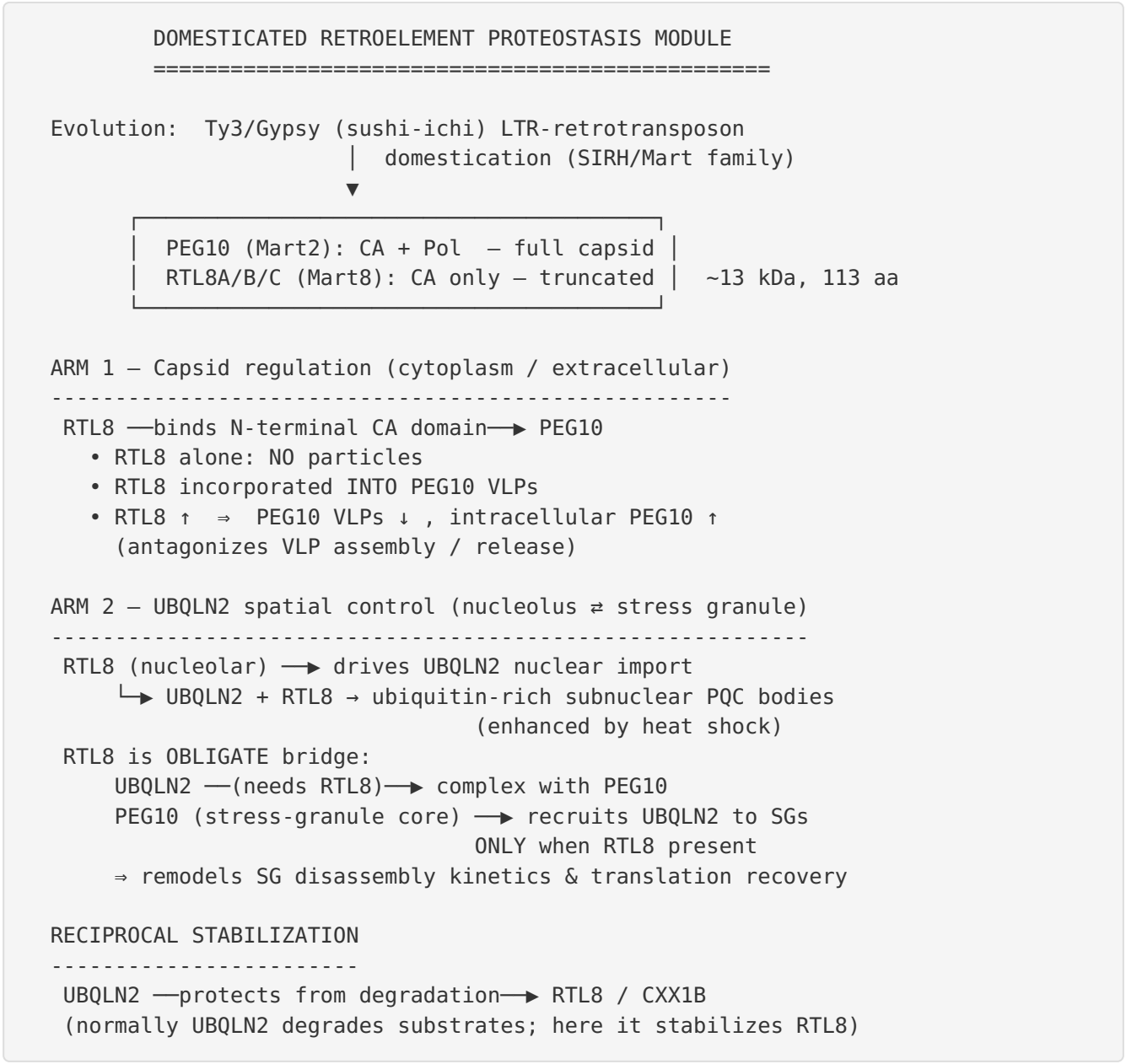
| Paralog pair | % identity | Substitutions |
|---|---|---|
| RTL8C vs RTL8A | 93.8% | 7 |
| RTL8C vs RTL8B | 92.9% | 8 |
| RTL8A vs RTL8B | 96.5% | 4 |

The differences are **few, scattered, and mostly conservative** (e.g., I8M, T20A, G48S, S60N). This very high identity is the formal justification for transferring experimental results obtained on "RTL8," RTL8A, or the CXX1B paralog onto RTL8C: the proteins share essentially the same fold and interaction surfaces and are expected to be functionally equivalent in the PEG10/UBQLN2 module.

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## Mechanistic Model / Interpretation

RTL8C is best understood as a **domesticated capsid-domain adaptor** that regulates two retroelement-derived proteostasis processes. The unifying theme is that RTL8, being a bare capsid (CA) module, acts by **docking onto other capsid or shuttle proteins and changing their behavior and location**, without any catalytic activity of its own.




**Primary function.** RTL8C is not an enzyme (no catalyzed reaction, no substrate turnover), not a transporter, and not a canonical signaling molecule. Its primary function is as a **protein-interaction adaptor built from a retroviral capsid domain**. Through this single CA surface it (a) binds the PEG10 capsid to **restrain VLP assembly**, and (b) binds UBQLN2 to **control its subcellular localization** and to **enable its coupling to PEG10-containing stress granules**.

**Localization.** RTL8C operates in **at least two compartments**: the **nucleolus** / **ubiquitin-enriched subnuclear PQC bodies** (where it drives UBQLN2 nuclear translocation) and the **cytoplasm** / **stress granules** (where it bridges UBQLN2 to PEG10). Because it is incorporated into PEG10 VLPs, RTL8C can also be present in / associated with **particle (extracellular-vesicle-like) compartments**.

**Pathways.** The relevant biochemical pathways are the **ubiquitin–proteasome system (UPS)** and **stress-granule-based proteostasis**, acting through the proteasomal shuttle factor UBQLN2, and the **endogenous-retroelement particle (VLP) pathway** centered on PEG10. RTL8C sits precisely at the junction of these two systems. This junction is medically significant: UBQLN2 mutations cause ALS/FTD, and PEG10 accumulation is implicated in neurodegeneration, so RTL8-mediated coupling of these factors is a candidate node in neurodegenerative proteostasis.

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# Evidence Base

| PMID | Title (abbreviated) | Contribution to the RTL8C model | Evidence type |
|---|---|---|---|
| 39775359 | *The gag-like gene RTL8 antagonizes PEG10-mediated virus-like particles* | RTL8 = truncated PEG10 CA domain; binds PEG10 N-terminal capsid; antagonizes VLP formation (Findings 1, 2) | Primary experimental (cell biology, biochemistry) |
| 35247097 | *RTL8 promotes nuclear localization of UBQLN2 to subnuclear compartments associated with PQC* | RTL8 nucleolar localization; drives UBQLN2 nuclear import to PQC bodies; UBQLN2-specific; reciprocal stabilization (Finding 3) | Primary experimental (MS screen, in vitro + in vivo validation) |
| 40680123 | *Endogenous retrovirus-like proteins recruit UBQLN2 to stress granules and shape their functional biology* | RTL8 is obligate bridge for UBQLN2–PEG10 complex and UBQLN2 recruitment to stress granules; remodels SG dynamics (Finding 4) | Primary experimental (cell biology) |
| 39484508 | *Endogenous retrovirus-like proteins recruit UBQLN2 to stress granules and alter their functional properties* | Companion report: RTL8 confers on UBQLN2 the ability to complex with/regulate PEG10; PEG10 forms VLPs within stress granules | Primary experimental |
| 33277362 | *Global proteomics of Ubqln2-based murine models of ALS* | Unbiased proteomics: PEG10 and CXX1B (RTL8 paralog) are top UBQLN2-linked proteins; UBQLN2 protects RTL8 from degradation (Finding 5) | Primary experimental (unbiased proteomics, orthogonal) |
| 25888968 | *Selective expression of sense/antisense transcripts of the SIRH family during mouse placentogenesis* | Places RTL8/Mart8 in the SIRH/Mart family of Ty3/gypsy LTR-retroelement-derived genes (Finding 1) | Evolutionary / expression |
| 34029402 | *Interactions between folded and disordered domains impart asymmetric effects on UBQLN2 phase separation* | Context: molecular basis of UBQLN2 LLPS (UBL/UBA + STI1 regions) that RTL8 modulates spatially | Primary biophysical (NMR) |

| PMID | Title (abbreviated) | Contribution to the RTL8C model | Evidence type |
|---|---|---|---|
| 29526694 | *Ubiquitin modulates LLPS of UBQLN2 via disruption of multivalent interactions* | Context: UBQLN2 phase separation and stress-granule recruitment switched by ubiquitin binding | Primary biophysical |
| 37501540 | *UBQLN2 and HSP70 participate in Parkin-mediated mitophagy* | Context: broader UBQLN2 proteostasis roles (not RTL8-specific) | Primary experimental |

**Strength of the evidence.** The RTL8C functional model rests on **convergent evidence from independent laboratories and methods**: targeted cell-biology/biochemistry (PMIDs 39775359, 35247097, 40680123/39484508), orthogonal unbiased proteomics (P 33277362), evolutionary classification (P 25888968), and structural inference from AlphaFold plus paralog sequence analysis (Finding 6). The two core claims — (1) RTL8 binds the PEG10 capsid and antagonizes VLP assembly, and (2) RTL8 is the obligate bridge coupling UBQLN2 to PEG10/stress granules — are each supported by direct, verifiable experimental statements with in vitro and in vivo validation. No contradictory literature was identified.

## Limitations and Knowledge Gaps

1. **Paralog resolution.** Most experiments describe "RTL8" collectively and do not distinguish RTL8A, RTL8B, and RTL8C individually. The >92% identity among paralogs (Finding 6) makes functional transfer to RTL8C reasonable, but **isoform-specific functions, expression differences, or non-redundancy of RTL8C specifically have not been directly tested**. Antibody and assay cross-reactivity across the cluster is a practical obstacle.

2. **Structural data are predicted, not experimental.** The capsid-like fold of RTL8C rests on an AlphaFold model (high confidence) and homology to PEG10, not on an experimental structure. **No crystal or cryo-EM structure of RTL8C alone or of an RTL8–PEG10 capsid complex** has been determined, so the precise capsid-interaction interface is inferred rather than observed.
3. **Mechanism of VLP antagonism is model-level.**

   P 39775359

   proposes that RTL8 "inhibits PEG10 VLP formation or release," but whether RTL8 blocks assembly, destabilizes assembled particles, or impairs secretion is not fully resolved.
4. **Physiological and tissue context.** The functional studies are largely cell-line based (with some in vivo mouse-brain validation for the UBQLN2 interaction). The **endogenous stoichiometry, tissue-specific roles (especially in human brain/neurons), and developmental functions of RTL8C** remain underexplored. No enzymatic, transporter, or receptor activity has been demonstrated — the "adaptor" assignment is by exclusion plus positive interaction data.
5. **Disease causality.** Links to ALS/FTD are through UBQLN2 and PEG10; whether RTL8C dysregulation is itself a driver, modifier, or bystander in neurodegeneration is not established.
6. **No catalytic/substrate characterization is possible** because the protein is not an enzyme or transporter; the "primary function" question is answered at the level of binding partners and regulatory outcome rather than a catalyzed reaction.

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## Proposed Follow-up Experiments / Actions

1. **Paralog-specific functional dissection.** Use CRISPR to individually knock out RTL8A, RTL8B, and RTL8C, and perform paralog-specific rescue, to determine whether RTL8C has non-redundant roles in PEG10 VLP antagonism and UBQLN2 localization.
2. **Structural determination of the RTL8–PEG10 capsid interface.** Solve a cryo-EM or crystal structure of RTL8C in complex with the PEG10 N-terminal capsid domain to define the interaction surface and test whether RTL8 acts as a chain-terminating "poison subunit" in capsid lattice assembly.

3. **Mapping the UBQLN2-binding determinant.** Use the UBQLN2-vs-UBQLN1/4 specificity (Finding 3) to identify the exact UBQLN2 region recognized by RTL8 (e.g., via domain-swap/mutagenesis and NMR/HDX), clarifying how RTL8 drives UBQLN2 nuclear import and stress-granule recruitment.
4. **Quantitative stress-granule dynamics.** Systematically titrate RTL8C, UBQLN2, and PEG10 and measure stress-granule assembly/disassembly kinetics and translation recovery, to quantify RTL8C's contribution to condensate remodeling.
5. **Mechanism of reciprocal stabilization.** Determine how UBQLN2 protects RTL8/CXX1 from degradation (Finding 5) — e.g., shielding from E3 ligases, altered ubiquitination, or sequestration — using cycloheximide chase and ubiquitination assays.
6. **Neuronal and disease-model context.** Examine RTL8C in human iPSC-derived neurons and in UBQLN2-mutant ALS/FTD models to test whether RTL8C modulates PEG10 accumulation and proteostasis in a disease-relevant setting.
7. **Endogenous localization and tissue atlas.** Generate paralog-discriminating reagents to map endogenous RTL8C localization (nucleolus vs. stress granule vs. VLP) across human tissues and stress states.

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## Conclusion

RTL8C (A6ZKI3; CXX1/FAM127A/Mart8C) is a small, domesticated Ty3/Gypsy-retroelement-derived protein consisting of a single Gag/capsid-like domain — a truncated homolog of PEG10. It is **not an enzyme, transporter, or classical signaling protein** but a **capsid-based protein-interaction adaptor** that (1) binds the PEG10 N-terminal capsid and antagonizes PEG10 virus-like-particle assembly/release, and (2) acts as the obligate bridge that drives UBQLN2 into nucleolar/subnuclear protein-quality-control compartments and couples UBQLN2 to PEG10-containing cytoplasmic stress granules. It therefore operates at the interface of **endogenous-retroelement particle biology and ubiquitin–proteasome/stress-granule proteostasis**, functioning in both the **nucleolus and the cytoplasm**, with notable relevance to UBQLN2/PEG10-linked neurodegeneration.

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